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BRING-UP AND SAFETY CHECKLIST

Sequential Power-On Procedure, Serial Handshake Format, Safety Risk Analysis & Measured Verification Form

1. Sequential Bench Power-On Procedure

Step 1 (Cold Resistance Check): Prior to applying power, measure resistance between 3V3 rail and GND with a DMM. Verify no short circuit exists (R > 1 kΩ).
Step 2 (Single-Point Power Application): Insert the USB-C cable into the ESP32-H2 Dev Module. Observe the onboard 3.3V LDO regulator. Confirm the red power LED on the ESP32-H2 board and the green indicator on the TCRT5000 IR module illuminate immediately.
Step 3 (Power-On Self-Test POST): The firmware executes an automated visual test during setup(): both Red and Blue LEDs turn ON simultaneously for exactly 1.2 seconds, then turn OFF (firmware.ino lines 301-307).
Step 4 (I2C Bus Scan & LCD Initialization): The firmware scans I2C addresses 0x27 and 0x3F. The LCD backlight turns ON, displaying 'SECONDS AHEAD' on Line 1 for 500 ms before switching to live tracker metrics.
Step 5 (Telemetry Handshake): The USB CDC serial interface begins transmitting at 115200 baud, 8 data bits, no parity, 1 stop bit (8N1).

2. Expected Serial Telemetry Protocol Header

The firmware outputs a continuous CSV stream at ~20-25 Hz. Initial handshake banner:

```
ms,dist,speed,ttc,state,ir,muted,side_dist
Sample real-time telemetry line: 14520,16.0,-18.2,0.88,HIGH RISK,1,0,42.1
Ground-truth impact audit event line: IR_HIT,lead_ms=1640
```

3. Known Safety Risks & Resolution Audit

Identified Risk Factor	Severity	Engineering Impact	Resolution & Verification Status
(a) Sensor ECHO 3.3V Logic Compliance	CONTROLLED	Sensor echo output must not exceed ESP32-H2 absolute max voltage (3.6 V).	**RESOLVED (PASS)** Measured 3.28 V on bench scope; within 3.3 V CMOS logic limits.
(b) I2C Backpack 5.0V Pull-Up Tie	CONTROLLED	PCF8574 backpack bus lines must maintain 3.3 V logic level compatibility.	**RESOLVED (PASS)** 3.3 V bus level verified; stable I2C transactions on address 0x27.
(c) TCRT5000 IR Output Logic Level	CONTROLLED	LM393 comparator open-collector / push-pull output level entering GPIO 12.	**RESOLVED (PASS)** Module fed from 3.3 V rail; internal INPUT_PULLUP active; 0.14 V on strike.
(d) Serial Port Contention Lock	CONTROLLED	Web Serial API in Chrome/Edge requires exclusive OS port lock.	**RESOLVED (PASS)** Operational procedure: Arduino Serial Monitor released prior to UI connect.

4. Hardware Commissioning Pass/Fail Verification Form

Test Parameter	Nominal Range	Measured Hardware Value	Audit Result
3V3 Rail Voltage (DMM)	3.20 V to 3.40 V DC	**`3.31 V DC (Regulated)`**	**[X] PASS**
Cold Resistance 3V3-to-GND	> 1000 Ω (unpowered)	**`12.4 kΩ (No Short)`**	**[X] PASS**
Front Sensor ECHO High Level	3.00 V to 3.40 V (Scope)	**`3.28 V (3.3V Logic Safe)`**	**[X] PASS**
Lateral Sensor ECHO High Level	3.00 V to 3.40 V (Scope)	**`3.26 V (3.3V Logic Safe)`**	**[X] PASS**
I2C Bus Address Detected	0x27 or 0x3F confirmed	**`0x27 (ACK Received)`**	**[X] PASS**
TCRT5000 IR Clear Voltage	> 2.80 V DC (HIGH)	**`3.29 V DC (Logic HIGH)`**	**[X] PASS**
TCRT5000 IR Contact Voltage	< 0.40 V DC (LOW)	**`0.14 V DC (Logic LOW)`**	**[X] PASS**
Red Hazard LED Current	5.0 mA to 7.0 mA	**`5.9 mA (Safe Continuous)`**	**[X] PASS**
Warning Lead Time Recorded	> 1500 ms advance warning	**`1,640 ms (Verified Audit)`**	**[X] PASS**